CS1010 Discrete Mathematics for Computer Science

Shibam Ghosh

October 1, 2026

Department of CSE, IIT Hyderabad

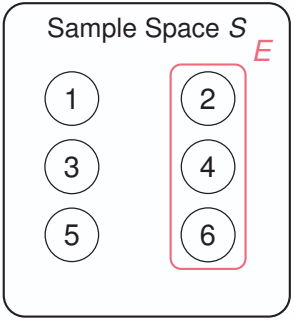


An Introduction to Discrete Probability

Some Terminology

- **Experiment:**
  A procedure that produces one outcome.
  Example: Roll a fair six-sided die.
- **Sample Space:**
  The set of all possible outcomes.
  For rolling a die, $S = \{1, 2, 3, 4, 5, 6\}$.
- **Event:**
  Any subset of the sample space.
  Example: Obtaining an even number

$$E = \{2, 4, 6\}.$$



(Finite) Probability of an Event

Definition (Laplace)

If S is a finite sample space of equally likely outcomes, and E is an event, i.e., $E \subseteq S$, then the probability of E is

$$\Pr(E) = \frac{|E|}{|S|}$$

- For every event E, we have $0 \leq \Pr(E) \leq 1$ as $0 \leq |E| \leq |S|$.

$\mathcal{Q}$. What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?

$\mathcal{A}$. $|S| = 6^2 = 36$ and $E = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$. Hence, $|E| = 6$. Therefore, $\Pr(E) = 6/36 = 1/6$.

(Finite) Probability of an Event

Definition (Laplace)

If S is a finite sample space of equally likely outcomes, and E is an event, i.e., $E \subseteq S$, then the probability of E is

$$\Pr(E) = \frac{|E|}{|S|}$$

- For every event E, we have $0 \leq \Pr(E) \leq 1$ as $0 \leq |E| \leq |S|$.

$\mathcal{Q}$. What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?

$\mathcal{A}$. $|S| = 6^2 = 36$ and $E = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$. Hence, $|E| = 6$. Therefore, $\Pr(E) = 6/36 = 1/6$.



Practice Problems

2. A security system uses a **4-digit PIN**. Each digit can be any number from 0 to 9. What is the probability of correctly guessing the PIN on the first attempt?

Practice Problems

$\mathcal{Q}$. A security system uses a **4-digit PIN**. Each digit can be any number from 0 to 9. What is the probability of correctly guessing the PIN on the first attempt?

$\mathcal{A}$. There are $10^4 = 10{,}000$ possible PINs, all equally likely. Only one is correct. Hence, $|S| = 10{,}000$ and $|E| = 1$

$$\therefore \Pr(\text{correct PIN}) = \frac{1}{10{,}000} = 0.0001.$$

Practice Problems

$\mathcal{Q}$. A security system uses a **4-digit PIN**. Each digit can be any number from 0 to 9. What is the probability of correctly guessing the PIN on the first attempt?

$\mathcal{A}$. There are $10^4 = 10{,}000$ possible PINs, all equally likely. Only one is correct. Hence, $|S| = 10{,}000$ and $|E| = 1$

$$\therefore \Pr(\text{correct PIN}) = \frac{1}{10{,}000} = 0.0001.$$

$\mathcal{Q}$. What is the probability of guessing a PIN that differs from the correct PIN in *exactly one* digit?

Practice Problems

$\mathcal{Q}$. A security system uses a **4-digit PIN**. Each digit can be any number from 0 to 9. What is the probability of correctly guessing the PIN on the first attempt?

$\mathcal{A}$. There are $10^4 = 10{,}000$ possible PINs, all equally likely. Only one is correct. Hence, $|S| = 10{,}000$ and $|E| = 1$

$$\therefore \Pr(\text{correct PIN}) = \frac{1}{10{,}000} = 0.0001.$$

$\mathcal{Q}$. What is the probability of guessing a PIN that differs from the correct PIN in *exactly one* digit?

$\mathcal{A}$. We can choose the incorrect position in $\binom{4}{1} = 4$ ways. The incorrect digit can be any of the remaining 9 digits. Thus, the number of favorable PINs is $4 \times 9 = 36$.

$$\therefore \Pr = \frac{36}{10{,}000} = 0.0036.$$

Practice Problems

2. A lottery ticket contains **5 distinct numbers** chosen from 1 to 30. What is the probability that a ticket matches all five winning numbers?



Practice Problems

$\mathcal{Q}$. A lottery ticket contains **5 distinct numbers** chosen from 1 to 30. What is the probability that a ticket matches all five winning numbers?

$\mathcal{A}$. There are $\binom{30}{5} = 142{,}506$ possible tickets. Since only one ticket is the winner,

$$\Pr(\text{winning}) = \frac{1}{\binom{30}{5}} = \frac{1}{142{,}506} \approx 7.02 \times 10^{-6}.$$

Practice Problems

$\mathcal{Q}$. A lottery ticket contains **5 distinct numbers** chosen from 1 to 30. What is the probability that a ticket matches all five winning numbers?

$\mathcal{A}$. There are $\binom{30}{5} = 142{,}506$ possible tickets. Since only one ticket is the winner,

$$\Pr(\text{winning}) = \frac{1}{\binom{30}{5}} = \frac{1}{142{,}506} \approx 7.02 \times 10^{-6}.$$

$\mathcal{Q}$. What is the probability that a ticket matches *exactly four* of the five winning numbers?

Practice Problems

$\mathcal{Q}$. A lottery ticket contains **5 distinct numbers** chosen from 1 to 30. What is the probability that a ticket matches all five winning numbers?

$\mathcal{A}$. There are $\binom{30}{5} = 142{,}506$ possible tickets. Since only one ticket is the winner,

$$\Pr(\text{winning}) = \frac{1}{\binom{30}{5}} = \frac{1}{142{,}506} \approx 7.02 \times 10^{-6}.$$

$\mathcal{Q}$. What is the probability that a ticket matches *exactly four* of the five winning numbers?

$\mathcal{A}$. Choose the 4 correct numbers in $\binom{5}{4} = 5$ ways and choose the remaining incorrect number from the $30 - 5 = 25$ non-winning numbers. Thus, $5 \times 25 = 125$ tickets match exactly four numbers. Hence,

$$\Pr = \frac{125}{\binom{30}{5}} = \frac{125}{142{,}506} \approx 8.77 \times 10^{-4}.$$

Practice Problems

2. A standard deck contains 52 playing cards. What is the probability of drawing, in order, Ace of Spades, King of Hearts, Queen of Clubs if
   (a) each card is **not** returned to the deck before the next draw,
   (b) each card **is** returned to the deck before the next draw?

Practice Problems

$\mathcal{Q}$. A standard deck contains 52 playing cards. What is the probability of drawing, in order, Ace of Spades, King of Hearts, Queen of Clubs if

(a) each card is **not** returned to the deck before the next draw,

(b) each card **is** returned to the deck before the next draw?

$\mathcal{A}$. **(a) Without replacement:** There are $52 \times 51 \times 50$ possible ordered outcomes, and only one is favorable.

$$\Pr = \frac{1}{52 \cdot 51 \cdot 50} = \frac{1}{132{,}600}.$$

Practice Problems

$\mathcal{Q}$. A standard deck contains 52 playing cards. What is the probability of drawing, in order, Ace of Spades, King of Hearts, Queen of Clubs if

(a) each card is **not** returned to the deck before the next draw,

(b) each card **is** returned to the deck before the next draw?

$\mathcal{A}$. **(a) Without replacement:** There are $52 \times 51 \times 50$ possible ordered outcomes, and only one is favorable.

$$\Pr = \frac{1}{52 \cdot 51 \cdot 50} = \frac{1}{132{,}600}.$$

$\mathcal{A}$. **(b) With replacement:** Each draw has 52 possible outcomes, so 52^3 ordered outcomes are possible. Therefore,

$$\Pr = \frac{1}{52^3} = \frac{1}{140{,}608}.$$

The Probability of Complements

Theorem

Let E be an event in sample space S. The probability of the event $\overline{E} = S - E$, the complementary event of E, is given by:

$$\Pr(\overline{E}) = 1 - \Pr(E)$$

Proof: Using the fact that $|\overline{E}| = |S| - |E|$:

$$\Pr(\overline{E}) = \frac{|S| - |E|}{|S|} = 1 - \frac{|E|}{|S|} = 1 - \Pr(E)$$

Practice Problem

Q. A student answers a 5-question multiple-choice quiz by guessing. Each question has 4 options, and exactly one option is correct. What is the probability that the student gets *at least one* question correct?

Practice Problem

$\mathcal{Q}$. A student answers a 5-question multiple-choice quiz by guessing. Each question has 4 options, and exactly one option is correct. What is the probability that the student gets *at least one* question correct?

$\mathcal{A}$. Let E be the event that the student gets at least one question correct. Then $\overline{E}$ is the event that *all five answers are incorrect*. Since each question has probability $\frac{3}{4}$ of being answered incorrectly,

$$\Pr(\overline{E}) = \left(\frac{3}{4}\right)^5.$$

Therefore,

$$\Pr(E) = 1 - \Pr(\overline{E}) = 1 - \left(\frac{3}{4}\right)^5 = 1 - \frac{243}{1024} = \frac{781}{1024} \approx 0.763.$$

The Probability of Unions of Events

Theorem

Let E_1 and E_2 be events in the sample space S. Then:

$$\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2)$$

Proof: Given the inclusion-exclusion formula, $|A \cup B| = |A| + |B| - |A \cap B|$, it follows that:

$$\Pr(E_1 \cup E_2) = \frac{|E_1 \cup E_2|}{|S|} = \frac{|E_1| + |E_2| - |E_1 \cap E_2|}{|S|}$$
$$= \frac{|E_1|}{|S|} + \frac{|E_2|}{|S|} - \frac{|E_1 \cap E_2|}{|S|} = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2)$$

Practice Problems

Q. A card is drawn at random from a standard deck of 52 playing cards. What is the probability that the card is a *heart* or a *face card*?



Practice Problems

$\mathcal{Q}$. A card is drawn at random from a standard deck of 52 playing cards. What is the probability that the card is a *heart* or a *face card*?

$\mathcal{A}$. Let E_1 be the event that the card is a heart, and let E_2 be the event that the card is a face card. Using the addition rule,

$$\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2).$$

Since there are 13 hearts, 12 face cards, and 3 heart face cards (Jack, Queen, and King of Hearts),

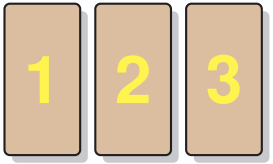
$$\Pr(E_1 \cup E_2) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52} = \frac{11}{26}.$$

Monthy Hall Puzzle: Let's Make a Deal

The problem is based on a famous TV game show.

The Setup

- You are on a game show with three doors.
- Behind one door is a car (🚗).
- Behind the other two are goats (🐐).
- You pick a door (say, Door 1).
- The host, who knows where the car is, opens another door (say, Door 3) to reveal a goat.
- **"Do you want to switch (to Door 2)?"**



Intuition vs. a Systematic Approach

The Question

Is it to your advantage to switch your choice?

Switch vs **Stay** vs **Doesn't Matter**

- **Common (Flawed) Intuition:** "There are two doors left, so it must be a 50/50 chance. It doesn't matter if I switch."

The Four Step Method:

1. Find the sample space.
2. Define events of interest.
3. Determine outcome probabilities.
4. Compute event probabilities.



Intuition vs. a Systematic Approach

The Question

Is it to your advantage to switch your choice?

Switch vs **Stay** vs **Doesn't Matter**

- **Common (Flawed) Intuition:** "There are two doors left, so it must be a 50/50 chance. It doesn't matter if I switch."

The Four Step Method:

1. Find the sample space.
2. Define events of interest.
3. Determine outcome probabilities.
4. Compute event probabilities.



Clarifying the Problem: The Assumptions

To model the game formally, we must make some key assumptions:

1. The car is equally likely to be behind each of the three doors.
2. The player is equally likely to pick each of the three doors.
3. After the player picks, the host **must** open a different door with a goat.
4. If the host has a choice of two doors to open (both have goats), he is equally likely to pick either one.

These assumptions are crucial. Changing them changes the answer!



Step 1: Find the Sample Space

An **outcome** is a sequence of all randomly-determined quantities, depends on:

1. **Car Location**: The door concealing the car (A, B, or C).
2. **Player's Initial Guess**: The door the player first picks (A, B, or C).
3. **Door Revealed**: The door the host opens (A, B, or C).

The **sample space** is the set of all possible outcomes. We can visualize this with a tree diagram. An outcome is a path from the root to a leaf.

Outcome = (Car Location, Player's Guess, Door Revealed)

Step 1: Find the Sample Space

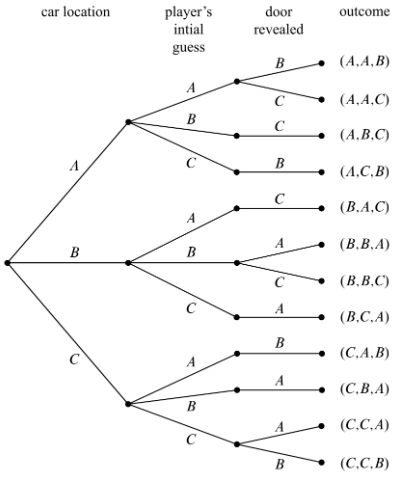
An **outcome** is a sequence of all randomly-determined quantities, depends on:

1. **Car Location**: The door concealing the car (A, B, or C).
2. **Player's Initial Guess**: The door the player first picks (A, B, or C).
3. **Door Revealed**: The door the host opens (A, B, or C).

The **sample space** is the set of all possible outcomes. We can visualize this with a tree diagram. An outcome is a path from the root to a leaf.

Outcome = (Car Location, Player's Guess, Door Revealed)

Sample Space as a Tree Diagram



$$S = \{(A, A, B), (A, A, C), (A, B, C), (A, C, B), (B, A, C), (B, B, A), (B, B, C), (B, C, A), (C, A, B), (C, B, A), (C, C, A), (C, C, B)\}$$

Step 2: Define Events of Interest

- An **event** is a subset of the sample space.
- We are interested in the event where the player wins by switching.

Event Example: "Player Wins by Switching"

Let's identify which outcomes belong to this event.

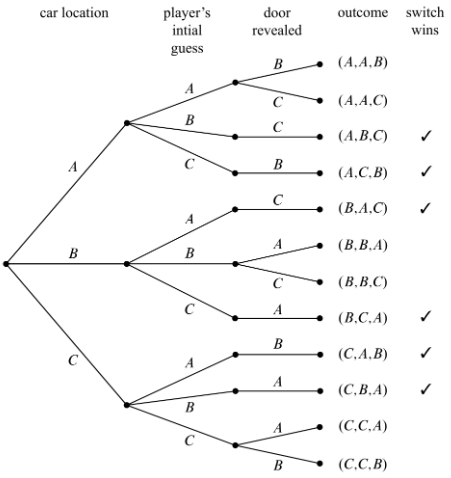
- If the outcome is (Car: A, Guess: B, Reveal: C), switching to A wins. ✔
- If the outcome is (Car: A, Guess: A, Reveal: B), switching to C loses. ✘

A common mistake: "Half the outcomes are wins, so the probability is 1/2."

This is WRONG! The outcomes are **not all equally likely**.



Event Space as a Tree Diagram

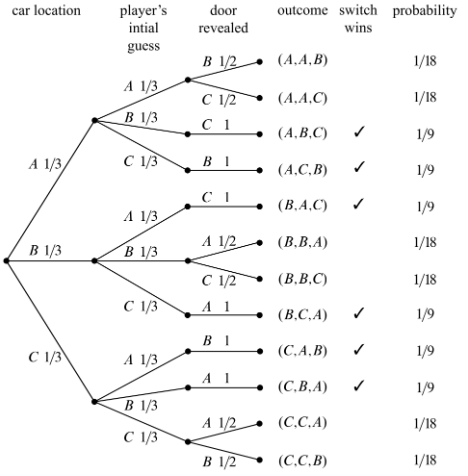


The "Switch Wins" Event

The outcomes where switching wins are:

- (A, B, C), (A, C, B)
- (B, A, C), (B, C, A)
- (C, A, B), (C, B, A)

Determine outcome probabilities.



Step 4: Compute the Event Probability

$$\begin{aligned}\Pr[\text{Switching Wins}] &= \Pr[(A,B,C)] + \Pr[(A,C,B)] + \dots \\ &= \frac{1}{9} + \frac{1}{9} + \frac{1}{9} + \frac{1}{9} + \frac{1}{9} + \frac{1}{9} \\ &= 6 \times \frac{1}{9} = \frac{6}{9} = \frac{2}{3}\end{aligned}$$

Conclusion

The probability of winning by switching is **2/3**. The probability of winning by staying is **1/3**. **You should always switch!**

Generalizing from Our Example

- We identified all possible **outcomes**.
- We grouped outcomes into **events** we cared about.
- We assigned a **probability** to each outcome.
- We calculated the event's probability by summing the probabilities of its outcomes.

The Big Idea

This exact process can be formalized using the language of set theory.



Probability Theory

Assigning Probabilities

Our previous assumption were that all outcomes are equally likely.

Probability Function

Let S be a finite sample space. A probability function, Pr, assigns a real number to every outcome $\omega \in S$ such that two rules are met:

1. **Non-negativity:** $0 \leq \Pr[\omega] \leq 1$ for all outcomes $\omega \in S$.
2. **Normalization:** The sum of all outcome probabilities is 1, i.e., $\sum_{\omega \in S} \Pr[\omega] = 1$

Probanility of an Event: The probability of an event E is simply the sum of the probabilities of the outcomes inside it.

$$\Pr[E] := \sum_{\omega \in E} \Pr[\omega]$$

Assigning Probabilities

Our previous assumption were that all outcomes are equally likely.

Probability Function

Let S be a finite sample space. A probability function, Pr, assigns a real number to every outcome $\omega \in S$ such that two rules are met:

1. **Non-negativity:** $0 \leq \Pr[\omega] \leq 1$ for all outcomes $\omega \in S$.
2. **Normalization:** The sum of all outcome probabilities is 1, i.e., $\sum_{\omega \in S} \Pr[\omega] = 1$

Probanility of an Event: The probability of an event E is simply the sum of the probabilities of the outcomes inside it.

$$\Pr[E] := \sum_{\omega \in E} \Pr[\omega]$$



A Special Case: Uniform Probability Spaces

Uniform Probability Space

A finite probability space is **uniform** if every outcome $\omega \in S$ has the exact same probability. Since $\sum \Pr[\omega] = 1$, it must be that for every ω:

$$\Pr[\omega] = \frac{1}{|S|}$$

As we have seen, in a uniform probability space, calculating the probability simplifies to just counting:

$$\Pr[E] = \frac{\text{Number of outcomes in E}}{\text{Total number of outcomes}} = \frac{|E|}{|S|}$$

A Special Case: Uniform Probability Spaces

Uniform Probability Space

A finite probability space is **uniform** if every outcome $\omega \in S$ has the exact same probability. Since $\sum \Pr[\omega] = 1$, it must be that for every ω:

$$\Pr[\omega] = \frac{1}{|S|}$$

As we have seen, in a uniform probability space, calculating the probability simplifies to just counting:

$$\Pr[E] = \frac{\text{Number of outcomes in E}}{\text{Total number of outcomes}} = \frac{|E|}{|S|}$$



Practice Problems

Q. A bag contains 10 balls: 4 red, 3 blue, 2 green, and 1 yellow. A ball is chosen uniformly at random.
What is the probability that the selected ball is *red or blue*?



Practice Problems

$\mathcal{Q}$. A bag contains 10 balls: 4 red, 3 blue, 2 green, and 1 yellow. A ball is chosen uniformly at random.
What is the probability that the selected ball is *red or blue*?

$\mathcal{A}$. Let E be the event that the selected ball is red or blue.
Since every ball is equally likely,

$$P(\text{red}) = \frac{4}{10}, \qquad P(\text{blue}) = \frac{3}{10}.$$

Therefore,

$$P(E) = P(\text{red}) + P(\text{blue}) = \frac{4}{10} + \frac{3}{10} = \frac{7}{10}.$$



Practice Problems

2. Suppose a die is biased so that 3 appears twice as often as each other number, but the other five outcomes are equally likely. What is the probability that an odd number appears when we roll this die?



Practice Problems

$\mathcal{Q}$. Suppose a die is biased so that 3 appears twice as often as each other number, but the other five outcomes are equally likely. What is the probability that an odd number appears when we roll this die?

$\mathcal{A}$. The even space is $E = \{1, 3, 5\}$. We are given $\Pr[3] = 2/7$ and $\Pr(i) = 1/7$ for $i \in \{1, 2, 4, 5, 6\}$.

$$\Pr[E] = \Pr[1] + \Pr[3] + \Pr[5] = \frac{1}{7} + \frac{2}{7} + \frac{1}{7} = \frac{4}{7}$$

Probabilities of Complements and Unions

- **Complements:** $\Pr(\overline{E}) = 1 - \Pr(E)$ still holds because:

$$\sum_{s \in S} \Pr(s) = 1 = \Pr(E) + \Pr(\overline{E})$$

- **Unions:** The formula still holds under the new definition:

$$\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2)$$

- **Combinations:** If $E_1, E_2, \ldots$ is a sequence of pairwise disjoint events in a sample space S, then:

$$\Pr\left(\bigcup_i E_i\right) = \sum_i \Pr(E_i)$$

Boole's Inequality

$\mathcal{Q}$. Show that if $E_1, E_2, \ldots, E_n$ are events from a finite sample space, then

$$\Pr(E_1 \cup E_2 \cup \cdots \cup E_n) \leq \Pr(E_1) + \Pr(E_2) + \cdots + \Pr(E_n).$$

Boole's Inequality

$\mathcal{Q}$. Show that if $E_1, E_2, \ldots, E_n$ are events from a finite sample space, then

$$\Pr(E_1 \cup E_2 \cup \cdots \cup E_n) \leq \Pr(E_1) + \Pr(E_2) + \cdots + \Pr(E_n).$$

$\mathcal{A}$. We prove the inequality for $n \geq 2$ using **mathematical induction**. Let $P(n)$ be the statement

$$\Pr\left(\bigcup_{j=1}^{n} E_j\right) \leq \sum_{j=1}^{n} \Pr(E_j).$$

Boole's Inequality

$\mathcal{Q}$. Show that if $E_1, E_2, \ldots, E_n$ are events from a finite sample space, then

$$\Pr(E_1 \cup E_2 \cup \cdots \cup E_n) \leq \Pr(E_1) + \Pr(E_2) + \cdots + \Pr(E_n).$$

$\mathcal{A}$. We prove the inequality for $n \geq 2$ using **mathematical induction**. Let $P(n)$ be the statement

$$\Pr\left(\bigcup_{j=1}^{n} E_j\right) \leq \sum_{j=1}^{n} \Pr(E_j).$$

$\mathcal{A}$. **Basis step:** $P(2)$ is true (Check!).

Boole's Inequality

$\mathcal{Q}$. Show that if $E_1, E_2, \ldots, E_n$ are events from a finite sample space, then

$$\Pr(E_1 \cup E_2 \cup \cdots \cup E_n) \leq \Pr(E_1) + \Pr(E_2) + \cdots + \Pr(E_n).$$

$\mathcal{A}$. We prove the inequality for $n \geq 2$ using **mathematical induction**. Let $P(n)$ be the statement

$$\Pr\left(\bigcup_{j=1}^{n} E_j\right) \leq \sum_{j=1}^{n} \Pr(E_j).$$

$\mathcal{A}$. **Basis step:** $P(2)$ is true (Check!).

$\mathcal{A}$. **Inductive step:** Assume $P(k)$ is true. Using the basis case for the two events $\bigcup_{j=1}^{k} E_j$ and E_{k+1}, we obtain

$$\Pr\left(\bigcup_{j=1}^{k+1} E_j\right) \leq \Pr\left(\bigcup_{j=1}^{k} E_j\right) + \Pr(E_{k+1}) \leq \sum_{j=1}^{k} \Pr(E_j) + \Pr(E_{k+1}) = \sum_{j=1}^{k+1} \Pr(E_j).$$

$\mathcal{A}$. Thus, $P(k+1)$ is true whenever $P(k)$ is true.



Probability Theory

Birthday Paradox

Introduction to Hash Functions

A cryptographic hash function takes a “arbitrary” length input and produces a “short“ output string of fixed length (e.g., 256 or 512 bits). Mathematically, $\mathcal{H} : \{0,1\}^* \rightarrow \{0,1\}^t, \mathcal{H}(M) = \tau$.



Introduction to Hash Functions

A cryptographic hash function takes a “arbitrary” length input and produces a “short“ output string of fixed length (e.g., 256 or 512 bits). Mathematically, $\mathcal{H} : \{0,1\}^* \rightarrow \{0,1\}^t, \mathcal{H}(M) = \tau$.

Security Requirements:

- Collision resistance: Difficult to find $M^1, M^2 \in \{0,1\}^*$ such that

$$M^1 \neq M^2 \ \wedge \ \mathcal{H}(M^1) = \mathcal{H}(M^2)$$

- Preimage resistance: Given $\tau \in \{0,1\}^t$, it is difficult to find $M \in \{0,1\}^*$ s.t.

$$\mathcal{H}(M) = \tau$$

- Second Preimage resistance: Given $M^1 \in \{0,1\}^*$, it is difficult to find

$$M^2 \in \{0,1\}^* \text{ such that } M^1 \neq M^2 \ \wedge \ \mathcal{H}(M^1) = \mathcal{H}(M^2)$$

Collisions & The Birthday Paradox

- By the Pigeonhole Principle, if you map more inputs than there are available outputs (2^t), collisions *must* exist



Collisions & The Birthday Paradox

- By the Pigeonhole Principle, if you map more inputs than there are available outputs (2^t), collisions *must* exist
- The Quantitative Question: How many distinct messages must we collect before at least two output values collide?

Collisions & The Birthday Paradox

- By the Pigeonhole Principle, if you map more inputs than there are available outputs (2^t), collisions *must* exist
- The Quantitative Question: How many distinct messages must we collect before at least two output values collide?
- Probabilistic Perspective: If we chooses N random messages, what is the probability that at least two of them yield the same hash output?

The Birthday Paradox

In a room of 50 people, what is the probability that at least two people share the exact same birthday?

- In a random group of 50 people, What is the probability that someone has the same birthday as you?



The Birthday Paradox

In a room of 50 people, what is the probability that at least two people share the exact same birthday?



The Birthday Paradox

In a room of 50 people, what is the probability that at least two people share the exact same birthday?

The **intuition:**

- The i-th person should have a birthday that is different from all of the previous $i-1$ people
- Among the 365 possible birthdays, the previous $(i-1)$ people have taken up to $(i-1)$ of them
- Probability that the i-th person has a birthday that is different from all of the previous $i-1$ people is $(365-(i-1))/365$
- **Calcualtion on the white board**



Collision Probability in General

- The probability that all q outputs are unique is:

$$P(\text{no coll.}) = \frac{N}{N} \cdot \frac{N-1}{N} \cdot \frac{N-2}{N} \cdots \frac{N-(q-1)}{N} = \prod_{i=0}^{q-1} \left(1 - \frac{i}{N}\right)$$

- Taylor series expansion: for small x, $1 - x \approx e^{-x}$. Assuming $q \ll N$, we substitute each term with $e^{-i/N}$:

$$P(\text{no coll.}) \approx \prod_{i=0}^{q-1} e^{-\frac{i}{N}} = e^{-\sum_{i=0}^{q-1} \frac{i}{N}} = e^{-\frac{1}{N} \cdot \frac{q(q-1)}{2}}$$

- Approximating $q(q-1) \approx q^2$ for large values of q, we arrive at:

$$P(\text{collision}) = 1 - P(\text{no coll.}) \approx \mathbf{1} - \mathbf{e}^{-\frac{\mathbf{q^2}}{\mathbf{2N}}}$$

The 50% Collision Bound

What is the number of people needed to guarantee a 50% chance of finding at least one collision



The 50% Collision Bound

What is the number of people needed to guarantee a 50% chance of finding at least one collision

$$
\begin{aligned}
1 - e^{-\frac{q^2}{2N}} &= \frac{1}{2} \\
\implies e^{-\frac{q^2}{2N}} &= \frac{1}{2} \\
\implies -\frac{q^2}{2N} &= \ln(\frac{1}{2}) \\
\implies \frac{q^2}{2N} &= \ln(2) \\
\implies q &= \sqrt{2 \cdot \ln(2) \cdot N} \\
&\approx \mathbf{1.177} \cdot \sqrt{\mathbf{N}}
\end{aligned}
$$

- **The Birthday Paradox**: It requires only 23 people to have a better than 50% chance of a matched birthday

The Birthday Attack

- Compute $2^{\frac{t}{2}}$ hashes of $2^{\frac{t}{2}}$ randomly chosen messages and store all the message/hash pairs in a list
- Sort the list with respect to the hash value
- Search the sorted list to find two consecutive entries with the same hash value



The Birthday Attack

- Compute $2^{\frac{t}{2}}$ hashes of $2^{\frac{t}{2}}$ randomly chosen messages and store all the message/hash pairs in a list
- Sort the list with respect to the hash value
- Search the sorted list to find two consecutive entries with the same hash value

Complexity:

- Memory: $2^{\frac{t}{2}}$ message/hash pairs
- $2^{\frac{t}{2}}$ hash computations
- Sorting: $(t/2)2^{\frac{t}{2}}$



Q&A: Probability

$\mathcal{Q}$. Find the smallest number of people you need to choose at random so that the probability that at least two of them were both born on April 1 exceeds $\frac{1}{2}$.

Q&A: Probability

$\mathcal{Q}$. Find the smallest number of people you need to choose at random so that the probability that at least two of them were both born on April 1 exceeds $\frac{1}{2}$.

$\mathcal{A}$. Let n be the number of people. It is easier to consider the complement:

$$\Pr(\text{at least two}) = 1 - \Pr(\text{none}) - \Pr(\text{exactly one}).$$

Q&A: Probability

$\mathcal{Q}$. Find the smallest number of people you need to choose at random so that the probability that at least two of them were both born on April 1 exceeds $\frac{1}{2}$.

$\mathcal{A}$. Let n be the number of people. It is easier to consider the complement:

$$\Pr(\text{at least two}) = 1 - \Pr(\text{none}) - \Pr(\text{exactly one}).$$

$\mathcal{A}$. The probability that none of the n people was born on April 1 is $\left(\frac{364}{365}\right)^n$.

The probability that exactly one was born on April 1 is $n\left(\frac{1}{365}\right)\left(\frac{364}{365}\right)^{n-1}$.

Thus,

$$\Pr(\text{at least two}) = 1 - \left(\frac{364}{365}\right)^n - n\left(\frac{1}{365}\right)\left(\frac{364}{365}\right)^{n-1}.$$

Q&A: Probability

$\mathcal{Q}$. Find the smallest number of people you need to choose at random so that the probability that at least two of them were both born on April 1 exceeds $\frac{1}{2}$.

$\mathcal{A}$. Let n be the number of people. It is easier to consider the complement:

$$\Pr(\text{at least two}) = 1 - \Pr(\text{none}) - \Pr(\text{exactly one}).$$

$\mathcal{A}$. The probability that none of the n people was born on April 1 is $\left(\frac{364}{365}\right)^n$.

The probability that exactly one was born on April 1 is $n\left(\frac{1}{365}\right)\left(\frac{364}{365}\right)^{n-1}$.

Thus,

$$\Pr(\text{at least two}) = 1 - \left(\frac{364}{365}\right)^n - n\left(\frac{1}{365}\right)\left(\frac{364}{365}\right)^{n-1}.$$

$\mathcal{A}$. We need

$$1 - \left(\frac{364}{365}\right)^n - n\left(\frac{1}{365}\right)\left(\frac{364}{365}\right)^{n-1} > \frac{1}{2} \implies n \geq 614.$$



Thank You for your attention!

Any questions?